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Application of Halbach magnetic assembly in the outer rotor of BLDC motor to improve its performance

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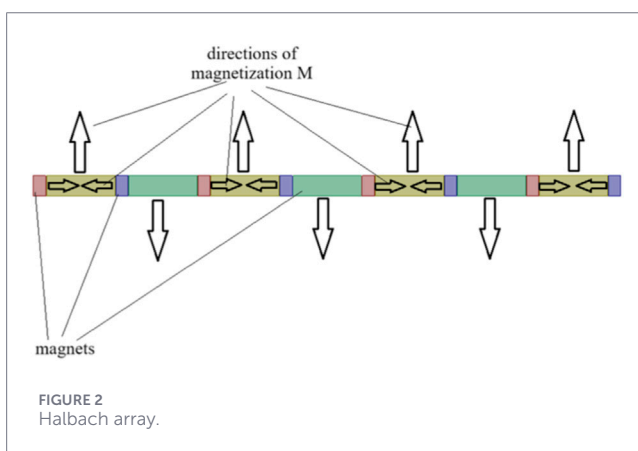
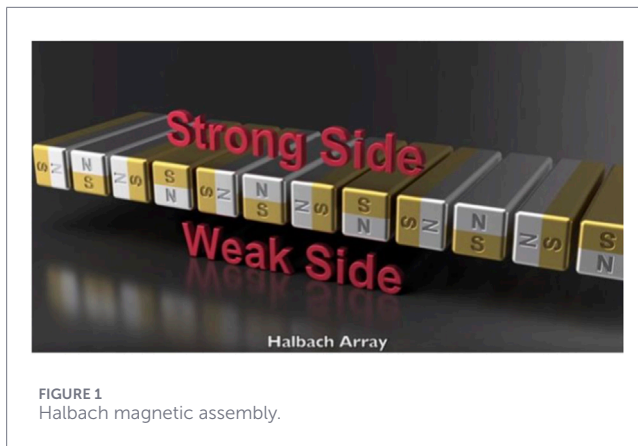
Modern electric vehicle drive systems are steadily moving toward simplified kinematics and improved energy efficiency. One promising solution is the use of hub motors, in which the electric motor is integrated directly into the vehicle wheel. Using external rotor motors as wheel hub motors is a technically sound and, in some cases, highly effective solution, especially for low- and medium-power electric vehicles. However, its effectiveness depends on a number of factors: design, operating mode, operating conditions, and drive system requirements. This article discusses the design of an external rotor BLDC using a Halbach magnet assembly, which will address the challenge of producing a highly efficient motor for vehicle hub motors. The motor was studied with various ratios of the outer rotor magnet sizes with radial and tangential magnetization vectors, and the optimal option for achieving the best motor performance was determined. It was demonstrated that for the 1000W motor under study, magnet mass can be reduced by 27% compared to a conventional motor. The magnetic field is concentrated in the working air gap, resulting in a 29% reduction in magnetic flux density in the rotor ring, which helps reduce magnetic losses and attenuate field leakage outside the magnetic circuit. The use of the Halbach magnetic system improves the motor's energy and performance characteristics: efficiency increases from 87.6% to 89.7%, and overload capacity increases by 6.4%, demonstrating increased energy conversion efficiency and improved dynamic performance of the electric machine.

KEYWORDS

current, efficiency, external rotor, Halbach magnetic assembly, hub motor, magnetic flux

1 Introduction

Currently, in-wheel motors are widely used in low-power vehicles. They feature a simplified kinematic design and high energy performance, such as efficiency, overload capacity, and starting torque. In modern in-wheel motors, the electric motor is



structurally integrated with the vehicle wheel (Chau, 2015; Zibarzani and Nasiri-Gheidari, 2026; Dai et al., 2025). This article presents the design methodology, optimization, and experimental validation of a high-efficiency and high-torque-density outer-rotor brushless DC (BLDC) motor developed for electric propulsion applications. A systematic, efficiency-oriented design approach is proposed, focusing on slot/tooth dimensioning and multistrand winding optimization to minimize copper and core losses. The optimized

motor was analyzed using finite element analysis (FEA). A wet-outer-rotor brushless DC motor with permanent magnets (WER-PMBLDC) is also featured, featuring an insulating sleeve separating the oil from the winding. This design improves winding reliability while eliminating dynamic sealing on the pump shaft. Partially submerged operation also helps balance flow resistance and heat dissipation.

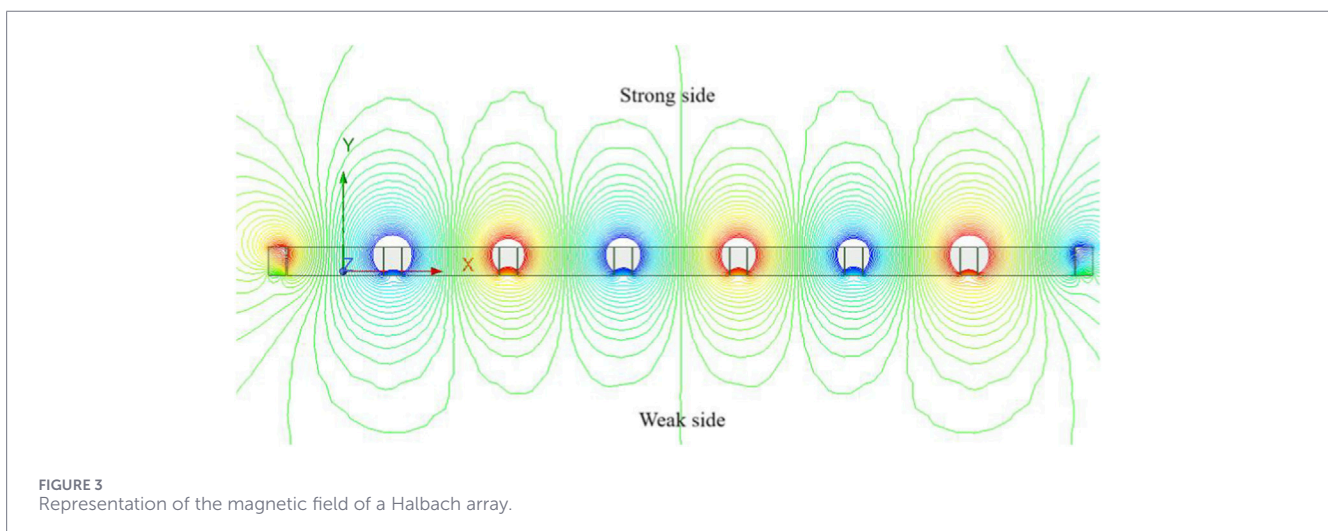
The use of external rotor motors in in-wheel hub motors is an effective solution offering a number of technical advantages, particularly for low- and medium-power electric vehicles. However, the effectiveness of this approach is determined by a number of factors, including the vehicle's design features, its operating modes, operating conditions, and the requirements placed on the drive system (Igorov et al., 2021c; Li et al., 2022).

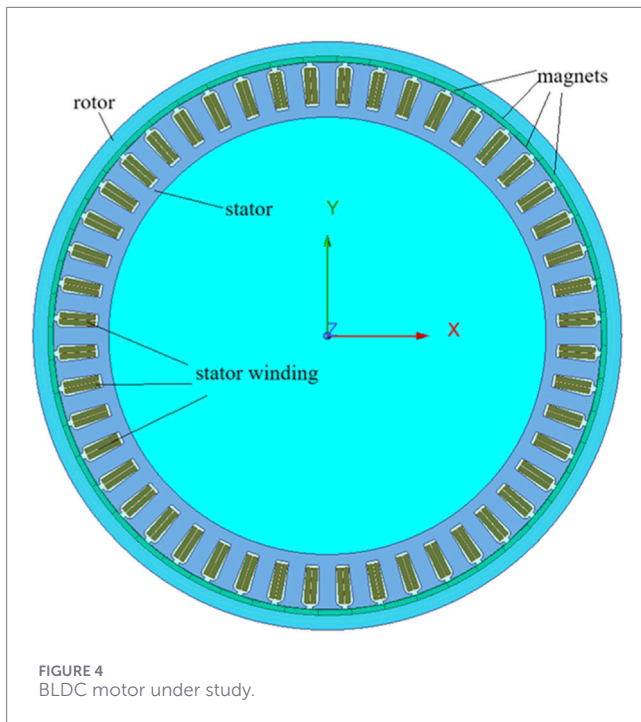
The methodological approach to solving this problem is based on integrating the electric motor with the vehicle's wheel and using an external rotor design. This approach increases the effective range of electromagnetic force application, allowing for increased torque without significantly increasing power consumption. Furthermore, eliminating intermediate mechanical gears helps reduce energy loss, improve drive reliability, and simplify vehicle design. Electromagnetic modeling, energy performance analysis, and comparative studies of various electrical machine designs are used to evaluate the effectiveness of this solution.

External rotor motors are characterized by the rotor being positioned around the stator. When used in in-wheel hub motors, this design solution offers several significant advantages.

- The large rotor diameter enables high torque, eliminating the need for a gearbox;
- There is no mechanical torque transmission, increasing overall system efficiency due to the elimination of gearbox and clutch losses.
- Good compactness is achieved by integrating the motor with the wheel, simplifying vehicle design.
- Energy recovery enables efficient braking and energy return to the battery.

External rotor hub motors are most effectively used in electric bicycles, electric scooters, light electric vehicles (urban vehicles),





and specialized equipment (robots, autonomous platforms) (Igorov et al., 2021a; Gadiyar et al., 2023).

Overall, external rotor motors are one of the most effective solutions for hub motors in low- and medium-power systems, particularly where compactness, high torque, and simple design are important.

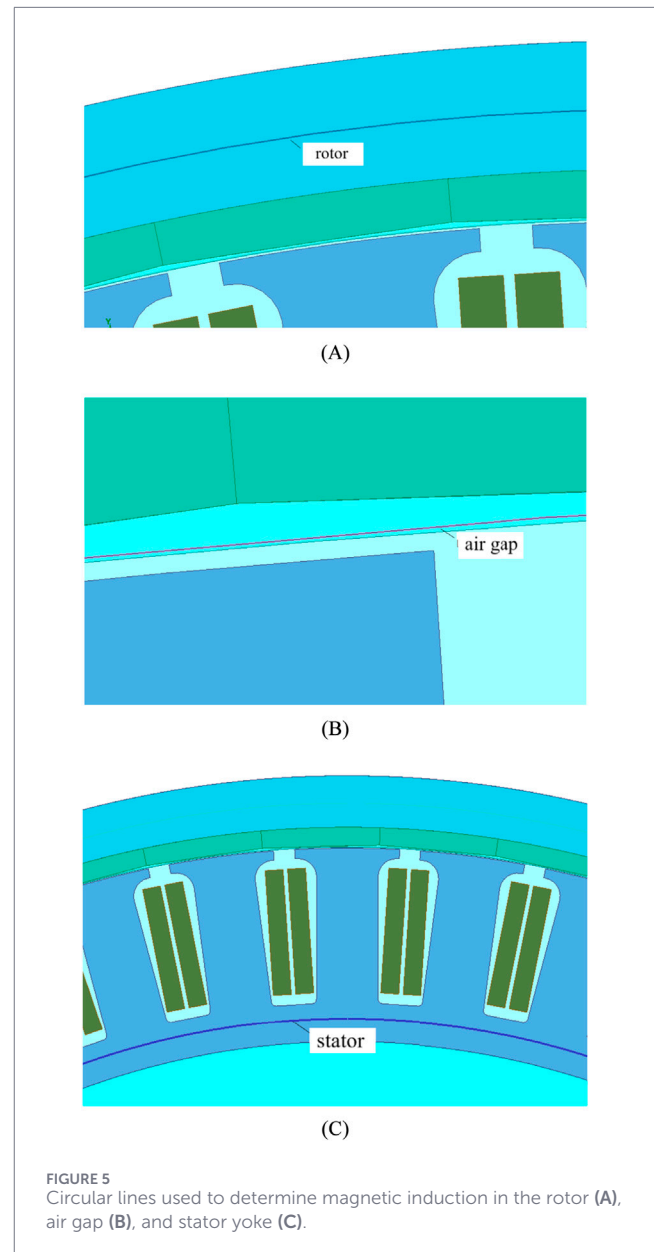
External rotor motors are typically based on brushless permanent magnet motors, as supported by modern research in electrical machines. In the hub motor design, the stator is rigidly mounted to the axle, while the rotor, aligned with the wheel rim, rotates around it (Wang et al., 2023; Li et al., 2023; Igorov et al., 2021b).

The key factor determining the efficiency of such motors is the increase in the radius (r) of application of electromagnetic forces (F), which leads to an increase in torque (Hanselman, 2012). Motor torque is determined by the (Equation 1):

$$M = F \times r \quad (1)$$

Increasing the rotor radius allows for high torque at relatively low currents, which positively impacts the drive's energy performance.

It is important for external rotor motors to achieve a high electromagnetic force (F) to achieve high torque within the compact design of the in-wheel motor. Based on this, the objective of this paper is to develop recommendations for the design of external rotor BLDC motors using a Halbach magnetic assembly, which will address the challenge of producing a highly efficient motor for vehicle in-wheel motors.



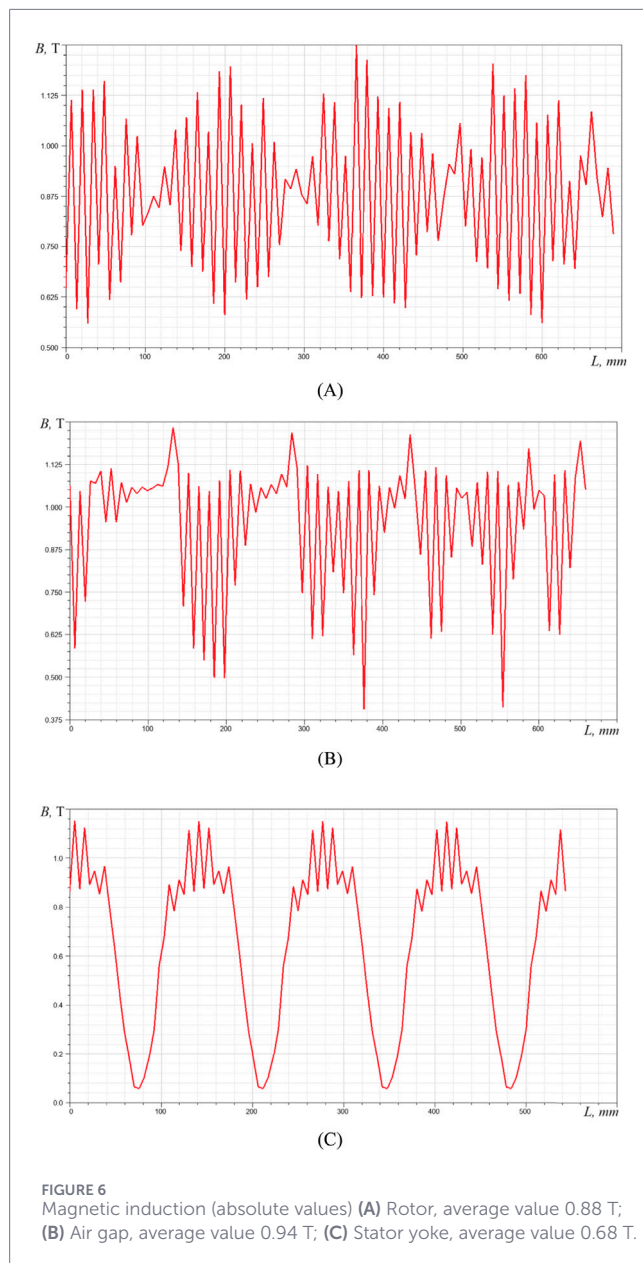
2 Materials and methods

The Halbach magnetic assembly (Figure 1) is a class of permanent magnetic systems in which, due to spatially rotating magnetization, a pronounced asymmetric distribution of the magnetic field is formed: on the “working” side, the field is enhanced, and on the opposite side, it is suppressed (Li et al., 2020).

For a linear array, this results in a harmonic field with exponential decay normal to the surface, while for a cylindrical configuration, it leads to a high flux concentration inside the bore or outside the rotor, depending on the polarity sign. Ideal analytical models describe the system well in the limit of continuous magnetization, infinite length, and $\mu_r \approx 1$. However, in real designs, discretization, higher harmonics, finite length, demagnetizing field, temperature stability, and manufacturing tolerances are crucial. Therefore, the practical engineering cycle for

TABLE 1 Parameters of the original BLDC motor based on test and calculation results.

Type of research	Mechanical power, W	Efficiency, %	Phase current, A	Rotor speed, rpm	Rotor torque, Nm
Tests	1000.0	87.6	15.5	450.0	21.2
Calculation	1000.0	88.7	15.3	450.0	21.2



Halbach arrays is almost always constructed as a combination of “analytical model - parametric optimization - FEM verification - experiment.” (Krause et al., 2002).

From a magnetostatics perspective, the physical formulation is based on Maxwell’s (Equation 2) in the medium.

$$\nabla \times \mathbf{H} = \mathbf{J}_f, \quad \nabla \cdot \mathbf{B} = 0, \quad \mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}), \quad (2)$$

which is equivalent (Equation 3)

$$\mathbf{H} = \frac{\mathbf{B}}{\mu_0} - \mathbf{M}. \quad (3)$$

If there are no free currents, then the source of the field is the magnetization $\mathbf{M}$ itself (Flanagan et al., 2025), distributed in the volume of the magnet (Figure 2).

The magnetized volume is equivalent to the volumetric and surface current densities, which are related to each other (Equation 4) (Björk et al., 2015; Meessen et al., 2010):

$$\mathbf{J}_m = \nabla \times \mathbf{M}, \quad \mathbf{K}_m = \mathbf{M} \times \mathbf{n}, \quad (4)$$

where $\mathbf{n}$ is the outward normal to the surface.

The objective of the study of a synchronous motor with a Halbach magnetic assembly on the rotor is to increase the magnetic induction in the air gap, increase the specific torque, and reduce the consumption of magnetic material. Parameters such as efficiency, stator winding current, and rotor torque were determined for the analysis.

To reduce computational costs, the following assumptions were made in the ANSYS program.

1. The magnetic circuit is made of homogeneous electrical steel.
2. The temperature of the permanent magnets is constant.
3. The air gap is the same along the entire perimeter of the machine.
4. The magnetic properties of the materials are isotropic.
5. The stator windings are symmetrical.
6. The influence of manufacturing tolerances is ignored.
7. Three-dimensional edge effects are initially replaced by a two-dimensional model.

During the study, the motor design was optimized by varying the magnet thickness; the number of Halbach segments; the stator tooth width; height of stator and rotor yoke.

This representation allows us to interpret the Halbach array as a specially organized set of current sheets that constructively sum on one side and destructively interfere on the other (Figure 3).

This effect of magnetic field enhancement on one side of the Halbach array and field weakening on the other is proposed to be used to improve the efficiency of external-rotor synchronous motors.

The study examined a 1000W BLDC motor with a 226 mm diameter external rotor. The rotor has 52 poles, generated by 12.8 mm wide magnets (Figure 4).

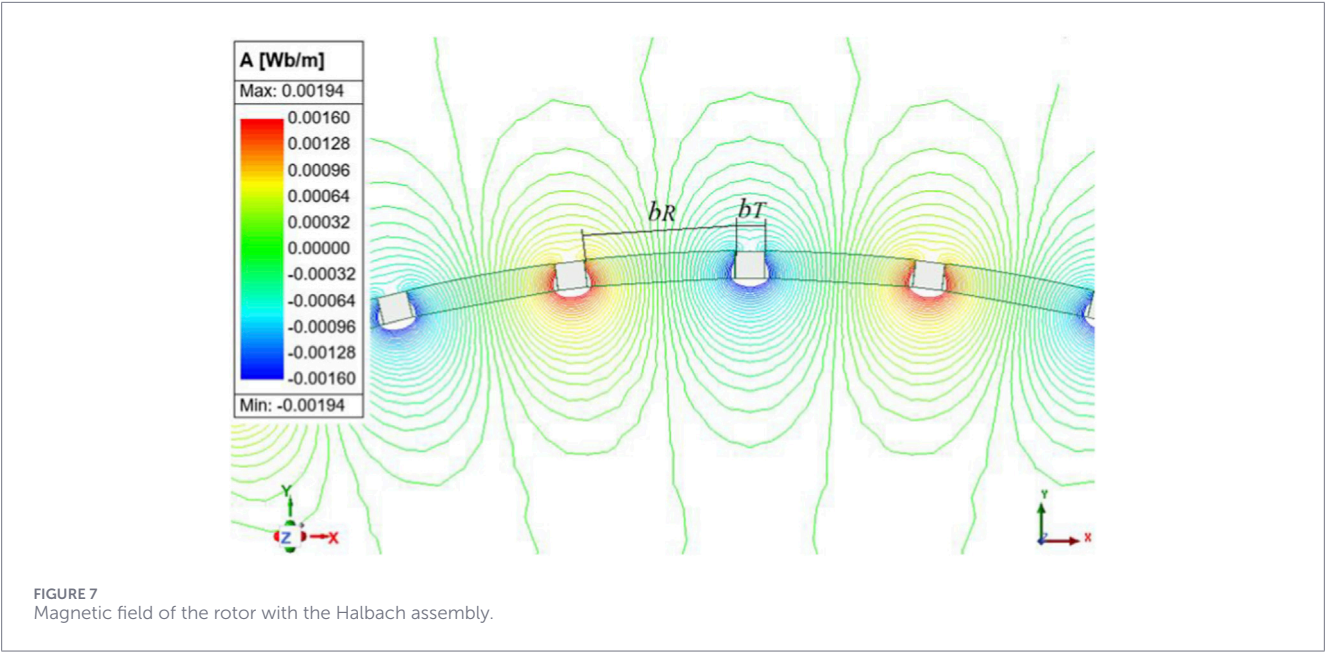


TABLE 2 Calculations with different magnet width ratios b_R/b_T .

No.	b_R/b_T , mm	B_r , T	B_d , T	B_s , T	I_0 , A	P_{Fe} , W
1	12.8/-	0.88	0.94	0.68	10.9	45.9
2	11.6/1.0	0.74	0.96	0.72	8.75	46.1
3	11.2/1.4	0.72	0.97	0.74	8.53	46.2
4	10.6/2.0	0.68	0.99	0.75	8.31	46.5
5	10.2/2.4	0.66	0.94	0.75	8.54	45.8
6	9.6/3.0	0.64	0.92	0.74	8.9	44.5
7	8.4/4.2	0.56	0.87	0.72	11.1	43.2
8	6.3/6.3	0.41	0.71	0.7	16.9	41.8

TABLE 3 Calculation results for motors with Halbach magnetic assembly and the original.

Motor type	b_R/b_T	Power, W	Efficiency, %	Phase current, A	Rotor speed, rpm	Rotor torque, Nm	Phase resistance, Ohm
Halbach assembly	10.6/2.0	1000.0	89.6	14.4	450.0	21.2	0.0815
Original	12.8/-	1000.0	88.7	15.3	450.0	21.2	0.0858

Ansys Maxwell software, which allows for the study of electromagnetic processes using the finite element method, was used for the calculations. This approach significantly improves the accuracy of the calculations (Özüpak, 2023; Chen and Zheng, 2024; Özüpak, 2022). At a nominal rotor speed of 450 rpm and a load torque of 21.2 Nm, the motor’s design parameters were determined, and the magnetic induction values in the rotor ring, air gap, and stator yoke were obtained. The circle lines on which these magnetic induction values were determined are shown in Figure 5.

3 Results and discussion

A number of scientific papers (Zuo et al., 2015; Boughrara et al., 2010; Xu et al., 2026; Li M. et al., 2026) devoted to the study of external-rotor synchronous motors have been reviewed. However, these papers do not provide a comparative analysis of motors with a Halbach assembly and a conventional rotor magnetic system design. Therefore, this article compares the performance of synchronous motors with a conventional rotor magnetic system

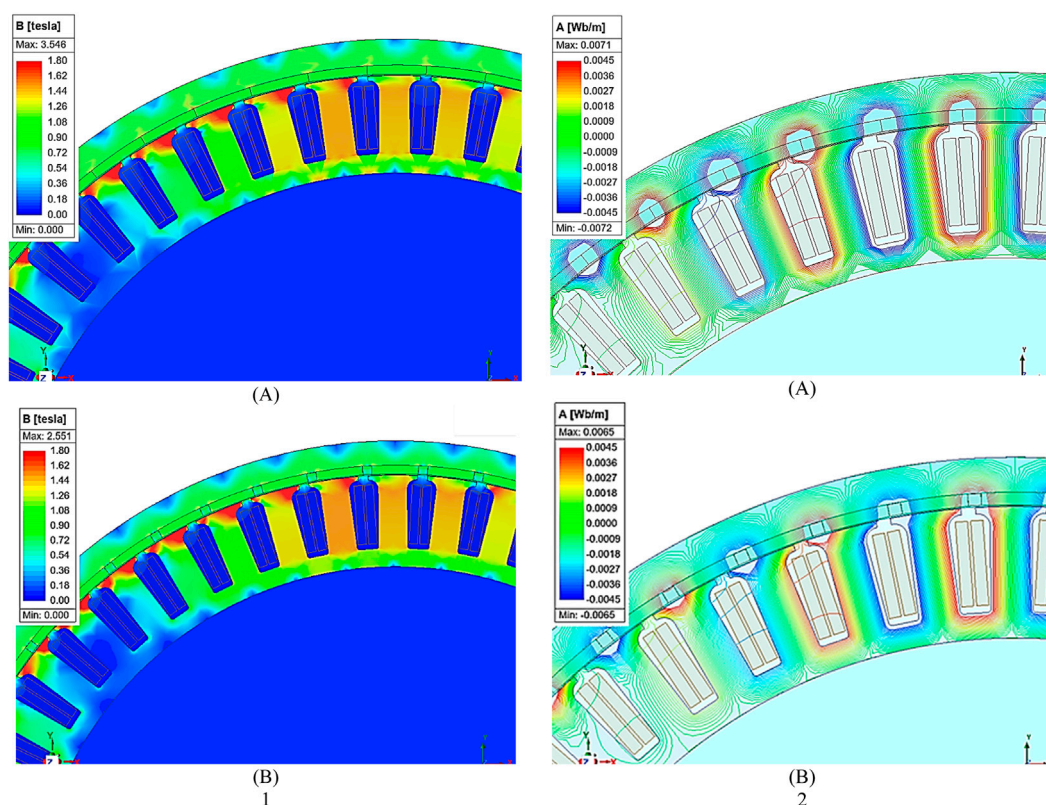


FIGURE 8
Magnetic induction values (1) and magnetic flux density (2) in the cross-section of the rotor and stator magnetic circuits (A) original motor (B) motor with Halbach assembly.



FIGURE 9
Rotor with Halbach magnetic assembly (option No. 4).

design and those with a Halbach assembly to demonstrate the effectiveness of the latter.

The calculation results were compared with motor test results to assess the validity of the developed BLDC motor model in Ansys Maxwell. Table 1 shows the calculated and experimental results.

Comparing the test and calculation results, we can conclude that the developed computational model in Ansys Maxwell adequately represents the electromagnetic processes of the BLDC motor under study.

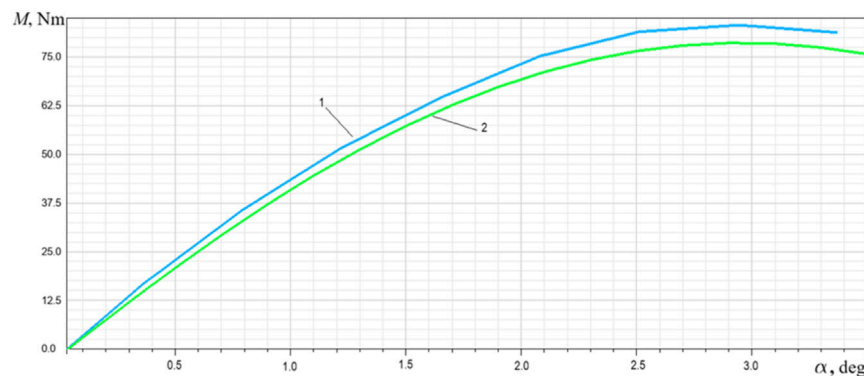
Calculated values of magnetic induction in idle mode were obtained for the circumferences, as shown in Figure 6: in the rotor ring, air gap, and stator yoke.

Next, a BLDC motor was simulated with a change in the magnetic design of the rotor, i.e., with the use of a Halbach magnetic assembly in order to increase the value of the magnetic field in the direction of the air gap and weaken the field towards the rotor ring (Figure 7).

An important factor when using Halbach magnetic assemblies is the ratio between the width of the main magnets (b_R), which create

TABLE 4 Test results for the base motor and the Halbach magnetic assembly.

Motor type	b_R/b_T	Mechanical power, W	Efficiency, %	Phase current, A	Rotor speed, rpm	Rotor torque, Nm
Halbach assembly	10.6/2.0	1000.0	89.7	14.2	450.0	21.2
Original	12.8/-	1000.0	87.6	15.5	450.0	21.2


FIGURE 10 Torque versus angle α . (1) motor with Halbach magnetic assembly (2) original motor.

a radially directed magnetic flux, and the width of the magnets (b_T), which create a tangentially directed magnetic flux (Figure 7). It is almost impossible to determine the b_R/b_T ratio analytically, since it is necessary to take into account the unevenness of the motor magnetic system, the effect of the magnetic flux created by the stator winding. Therefore, the studies were carried out in the Ansys Maxwell program using the finite element method. The following magnet width ratios b_R/b_T were selected for the study: 11.6 mm/1.0 mm; 11.2 mm/1.4 mm; 10.6 mm/2.0 mm; 10.2 mm/2.4 mm; 9.6 mm/3.0 mm; 8.4 mm/4.2 mm; 6.3 mm/6.3 mm. For each magnet width ratio b_R/b_T , a motor simulation model was created in Ansys Maxwell and run in idle mode with a rotor speed of 450 rpm. The following parameters were determined: magnetic induction in the rotor ring (B_r), air gap (B_s , B_δ), and stator yoke (B_s); stator winding current (I_0); and magnetic core iron losses (P_{Fe}). The results are presented in Table 2.

Analyzing the results in Table 2, we can conclude that the b_R/b_T ratio of 10.6 mm/2.0 mm (option No. 4) yields the lowest no-load current with the highest air gap flux density. Further increasing the width of the magnets, which create a tangentially directed magnetic flux, leads to a decrease in the air gap flux density and, consequently, an increase in the no-load current (I_0).

The nominal operating mode of the motor with rotor magnets for option #4 was calculated using Ansys Maxwell. It is important to take into account the fact that the air gap magnetic density increased by 5.3%. Therefore, the stator winding was adjusted to account for the increase in B_δ by changing the effective number of turns of the stator winding. The calculation results are presented in Table 3.

A comparative analysis of the induction values over the entire cross-sectional area of the magnetic circuit of the

original motor and the motor with the Halbach magnetic assembly was carried out (Figure 8).

Calculations showed that the magnetic inductions in the motor with the Halbach assembly (Figures 8, 1B) remain within acceptable limits. Compared to the original motor, the maximum inductions at individual points decreased from 3.546 T to 2.551 T, which is a positive change. A reduction in inductions in the rotor ring of the motor with the Halbach assembly is also evident. The magnetic flux density lines show that in the Halbach assembly motor the lines are concentrated closer to the air gap (Figures 8, 2B), which means a higher degree of utilization of the magnetic flux in generating rotor torque.

A prototype of a motor with a Halbach magnetic assembly was manufactured according to the size option of magnets No. 4 in Table 2 (Figure 9) and with the adjustment of the effective turns of the stator winding.

The motor was tested, and the results are compared with the test results of the base motor in Table 4.

It's worth noting the increased motor efficiency due to the reduced current in the stator winding. Iron losses in the magnetic circuit remain virtually unchanged when switching to a Halbach magnetic assembly.

Also, calculations determined that to achieve an air gap induction of 0.99 T with a conventional rotor magnetic system, the magnet mass would have to be increased by 27.2%. For the motor under study, this would mean an increase in magnet mass from 0.246 kg to 0.313 kg.

The use of a Halbach magnetic assembly allows for a reduction in rotor ring induction from 0.88 T to 0.68 T. This allows for a reduction in ring thickness without degrading motor performance, thereby reducing its weight.

A study was conducted on the overload capacity of a motor with a Halbach magnetic assembly. This indicator is important for vehicle motors, which must maintain operability under short-term overloads during movement: climbing an inclined surface, driving over an obstacle, *etc.* For this purpose, the motor torque was determined as the angle (α) between the rotor's geometric axis and the voltage vector changed, according to (Li X. et al., 2026; Kalender et al., 2011). The calculations were carried out in the range from idle mode, i.e., when the torque is 0, to maximum load mode (Figure 10).

It's clear that the motor with the Halbach magnet assembly on the rotor has a higher overload capacity than a conventional motor. The maximum torque at a rotor speed of 450 rpm for the motor with the Halbach magnet assembly is 83 Nm, compared to 78 Nm for the motor with the traditional rotor magnet arrangement.

4 Conclusion

The practical application of Halbach magnet assembly motors in BLDC electric drives leads to a number of significant engineering benefits: The mass and volume of the magnetic system are reduced while maintaining or improving output characteristics, which is particularly important for mobile and transport applications (in-wheel motors, unmanned platforms, robotics); cooling system requirements are reduced by reducing losses in the rotor section, simplifying the housing design and increasing reliability; and the motor's power density increases, enabling the design of more compact drives with the same traction characteristics. Furthermore, increased overload capacity improves the motor's resistance to dynamic conditions, such as starting under load, sudden torque changes, and regenerative braking.

Research has shown that the magnet mass required to create magnetic induction in the air gap is 27.2% less than that required for a similar synchronous motor of conventional design.

Redistributing magnetic flux density also reduces its value in the rotor ring by 29%, reducing losses and field dissipation outside the magnetic circuit. The Halbach rotor magnetic assembly increases motor efficiency from 87.6% to 89.7% and increases overload capacity by 6.4%.

Research into Halbach magnetic assembly technology requires further development in a number of areas: optimization of Halbach array segmentation; study of the influence of the number of segments and magnetization angles on efficiency, torque ripple, and magnetic loss; three-dimensional effects and edge zones; in-depth analysis of end-face effects in 3D models, especially for short machines and in-wheel motors.

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Data availability statement

The original contributions presented in the study are included in the article/supplementary material, further inquiries can be directed to the corresponding authors.

Author contributions

HA_I: Conceptualization, Writing – original draft, Methodology. SA-E: Software, Writing – review and editing. MQ: Writing – original draft, Investigation. OI (fourth author): Conceptualization, Writing – original draft. OI (fifth author): Writing – original draft, Project administration, Data curation. MG: Writing – original draft, Software, Validation. OM: Writing – review and editing, Formal Analysis.

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